

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

AI Research Engineer

AI Research Engineer with experience in AI evaluation, ML reliability and production model failure analysis. Proven experience using Python to build AI solutions for John Deere, British Petroleum and Austin Industries. I have worked on understanding why AI systems fail and how to make those patterns clear enough for engineering teams to act on. Experienced in benchmarking, post-deployment validation, and observability work before it becomes a customer issue. Comfortable turning messy performance signals into clear recommendations for product and engineering decisions.

PROJECTS

FinanceBench RAG Evaluation Harness

Jun 2026

- Built an evaluation framework for FinanceBench, a financial question-answering benchmark, to measure how well RAG and document-grounded QA systems answer questions using the right source evidence.
- Designed structured experiments across closed-book, oracle-context, and dense RAG workflows to identify whether failures came from retrieval, source selection, or answer generation.
- Added LLM-as-judge scoring, evidence matching, citation checks, numeric matching, and hallucination labels to make AI agent performance easier to inspect, compare, and improve.
- Produced reproducible runs using automated tests, config-driven experiments, structured prediction outputs, and Markdown reports, enabling regression checks and clearer technical recommendations for improving retrieval and answer quality.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end AI pipeline for real-time perception in harsh industrial environments, covering data curation, model training through deployment and on-site validation, improving F1 by 15% while maintaining deterministic 24/7 operation on Bristish Petroleum rigs.
- Built multimodal data ingestion, cleansing, training, and validation pipelines across image, video, and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records and reducing model rollout time from 3 days to under 24 hours.
- Applied statistical and experimental methods including Bayesian optimisation, ablation studies, hypothesis testing, and confidence intervals to support model selection, threshold tuning, and failure analysis in production.
- Led on-site validation and diagnostics on deployments, troubleshooting model failures under environmental stressors and translating performance findings into fixes that improved robustness in live operation.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, structured logging, health checks, retries, and safe restarts reducing live video pipeline crashes by 67% and improved root-cause analysis.
- Presented model performance, failure modes, and deployment trade-offs to engineering teams, client stakeholders, and field operators, turning messy production signals into clear optimisation recommendations.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety applications, reducing false positives by 33% while meeting latency and power constraints for Austin Industries and John Deere.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using Python and OpenCV.

- Built automated labelling, dataset versioning, and performance-tracking workflows, repurposing existing detection models for annotation and cutting manual labelling effort by 80% on large real-world datasets.
- Ran systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.
- Worked with hardware, software, QA, and client teams to translate model performance, latency, and failure analysis into rollout decisions for deployed safety systems.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Dissertation: designed and implemented a Radiological SLAM system on a Clearpath Husky platform, integrating LiDAR and radiological data for localisation, mapping, and inspection workflows with Python.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

CERTIFICATIONS

Building RAG Agents with LLMs

Nvidia

Jun 2026

Deep Learning Specialisation

deeplearning.ai

Jul 2019

SKILLS

- **AI Evaluation:** LLM-as-judge, RAG evaluation, document-grounded QA, retrieval metrics, evidence matching, citation checks, hallucination detection, error analysis, structured outputs
- **Experimentation and Reliability:** failure-mode analysis, benchmarking, regression checks, hypothesis testing, ablation studies, post-deployment validation, production monitoring
- **Python and ML Engineering:** Python, Pandas, NumPy, scikit-learn, PyTorch, MLflow, DVC, Apache Airflow, Optuna
- **Production Systems:** Docker, CI/CD, Git, structured logging, observability, health checks, Apache Kafka, Linux
- **Applied AI:** information extraction, anomaly detection, object detection, OpenCV, YOLO, OpenVINO, ONNX